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BICC1 Interacts with PKD1 and PKD2 to Drive Cystogenesis in ADPKD

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eLife Assessment

This study presented **valuable** findings regarding the basic molecular pathways leading to the cystogenesis of Autosomal Dominant Polycystic Kidney Disease, suggesting BICC1 functions as both a minor causative gene for PKD and a modifier of PKD severity. Although some **solid** data were supplied to demonstrate the functional and structural interactions between BICC-1, PC1 and PC2, respectively. The characterization of such interactions appear to be **incomplete**, which renders the specific relevance of these findings for the etiology of relevant diseases unclear.

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Abstract

Autosomal dominant polycystic kidney disease (ADPKD) is primarily of adult-onset and caused by pathogenic variants in *PKD1* or *PKD2*. Yet, disease expression is highly variable and includes very early-onset PKD presentations *in utero* or infancy. In animal models, the RNA-binding molecule Bicc1 has been shown to play a crucial role in the pathogenesis of PKD. To study the interaction between BICC1, PKD1 and PKD2 we combined biochemical approaches, knockout studies in mice and *Xenopus*, genetic engineered human kidney cells carrying BICC1 variants as well as genetic association studies in a large ADPKD cohort. We first demonstrated that BICC1 physically binds to the proteins Polycystin-1 and -2 encoded by *PKD1* and *PKD2* via distinct protein domains. Furthermore, PKD was aggravated in loss-of-function studies in *Xenopus* and mouse models resulting in more severe disease when *Bicc1* was depleted in conjunction with *Pkd1 or Pkd2*. Finally, in a large human patient cohort, we identified a sibling pair with a homozygous *BICC1* variant and patients with very early onset PKD (VEO-PKD) that exhibited compound heterozygosity of *BICC1* in conjunction with *PKD1 and PKD2* variants. Genome editing demonstrated that these *BICC1* variants were hypomorphic in nature and impacted disease-relevant signaling pathways. These findings support the hypothesis that BICC1 cooperates functionally with PKD1 and PKD2, and that *BICC1* variants may aggravate PKD severity highlighting RNA metabolism as an important new concept for disease modification in ADPKD.

Introduction

Autosomal dominant polycystic kidney disease (ADPKD) is the most frequent life-threatening genetic disease and one of the most common Mendelian human disorders with an estimated prevalence of 1/400-1000.[1,2] This equates to around 12.5 million affected individuals worldwide.

About 5-10% of all patients requiring renal replacement therapy are affected by ADPKD. The majority of ADPKD patients carry a pathogenic germline variant in the *PKD1* or *PKD2* gene and present with the disease in adulthood.[2–4] However, occasionally, ADPKD can manifest in infancy or early childhood [< 2 years, very-early onset ADPKD (VEO-ADPKD)], and in late childhood or early teenage years [2-16 years, early-onset ADPKD (EO-ADPKD)].[5,6] VEO patients and fetuses often present with a Potter sequence and significant peri- or neonatal demise, which can be clinically indistinguishable from a typical Autosomal Recessive Polycystic Kidney Disease (ARPKD) presentation caused by *PKHD1* mutations.[7,8] However, in contrast to VEO/EO-ADPKD, ARPKD kidneys invariably manifest as fusiform dilations of renal collecting ducts and distal tubules that usually remain in contact with the urinary system.[4] Co-inheritance of an inactivating *PKD1* or *PKD2* mutation with an incompletely penetrant minor PKD allele *in trans* provides a likely explanation for VEO-ADPKD.[9] In fact, we recently reported that the majority (70%) of VEO-ADPKD cases in an international diagnostic cohort had biallelic *PKD1* variants (i.e., a pathogenic variant *in trans* with a hypomorphic, low penetrance variant), while cases of biallelic *PKD2* and digenic *PKD1/PKD2* were rather rare.[10] In line with the dosage theory for PKD,[11] several other genes (*GANAB, DNAJB11, ALG8, ALG9, IFT140*) have been associated with a dominant, but late-onset atypical adult presentation and sometimes incomplete penetrance.[4,12–15] However, not all VEO/EO-ADPKD patients can be explained by monogenic inheritance suggesting digenic or oligogenic inheritance causes.

Previous data from mouse, *Xenopus* and zebrafish suggest a crucial role for the RNA-binding protein Bicc1 in the pathogenesis of PKD, although *BICC1* mutations in human PKD have not been previously reported.[16–25] BICC1 encodes an evolutionarily conserved protein that is characterized by 3 K-homology (KH) and 2 KH-like (KHL) RNA-binding domains at the N-terminus and a SAM domain at the C-terminus, which are separated by a by a disordered intervening sequence (IVS).[26–31] The protein localizes to cytoplasmic foci involved in RNA metabolism and has been shown to regulate the expression of several genes such as *Pkd2, Adcyd6* and *Pkia* in the kidney.[22,32] We now present data providing a mechanistic model linking BICC1 with the three major cystic proteins. We show that BICC1 physically interacts with the PKD1 (PC1) and the PKD2 (PC2) proteins in human kidney cells. We also demonstrate that *Pkd1* and *Pkd2* modifies the cystic phenotype in *Bicc1* mice in a dose-dependent manner and that Bicc1 functionally interacts with Pkd1, Pkd2 and Pkhd1 in the pronephros of *Xenopus* embryos. Finally, this interaction is supported by human patient data where *BICC1* alone or in conjunction with *PKD1* or *PKD2* is involved in VEO-PKD.

Methods

Cell Culture and Biochemical Studies

The characterization of the interaction between BICC1, PC1 and PC2 as well as the analysis of the human *BICC1* variants were performed using standard approaches detailed in the Supplementary Methods.

Animal Studies

Mouse and *Xenopus laevis* studies were approved by the Institutional Animal Care and Use Committee at the Cleveland Clinic Foundation and LSU Health Sciences Center (present and former employer of Dr. Wessely) and adhered to the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Experimental design and data interpretation followed the ARRIVE1 reporting guidelines.[33]

International Diagnostic Clinical Cohort

Research was performed following written informed consent and according to the declaration of Helsinki and oversight was provided by the Medizinische Genetik Mainz. DNA extraction was performed according to standard procedures (see Supplementary Methods for details).

Statistical analysis

Data are presented as mean values ± SEM. Paired and unpaired two-sided Student's t test or ANOVA were used for statistical analyses with a minimum of $P < 0.05$ indicating statistical significance.

Measurements were taken from distinct biological samples. Analyses were carried out using Prism 10 (Graphpad).

Results

Interaction of BICC1 with PC1 and PC2

Loss of Pkd1 has been associated with lower Bicc1 expression in a murine model.[34] Furthermore, Bicc1 has been shown to regulate *Pkd2* expression in cellular and animal models.[22,35,36] However, whether this is due to direct protein-protein interactions between BICC1, PKD1 (PC1) and PKD2 protein (PC2) have not been investigated. In pilot experiments, BICC1 was detected by mass spectrometry in a pulldown assay from cells stably expressing a Polycystin-1 PLAT domain (Polycystin-1, Lipoxygenase, Alpha-Toxin)-YFP fusion.[37] The direct binding between the PC1-PLAT domain and Bicc1 was confirmed using *in vitro* binding assays, but we also detected binding to the PC1 C-terminus (CT1) (Supplementary Fig. S1a,c).

Utilizing recombinant GST-tagged domains of PC1 and PC2, we demonstrated that Bicc1 binds to both proteins in GST-pulldown assays (Fig. 1a, b). In the case of PC1, myc-mBicc1 strongly interacted with its C-terminus (GST-CT1) but its interaction was abolished by a PC1-R4227X truncation mutation (GST-CT1-R4227X) (Fig. 1b,c). In the case of PC2, myc-mBicc1 associated with both recombinant GST N-terminal (GST-NT2) and C-terminal (GST-CT2) fusions. To investigate whether binding was direct or indirect, we performed *in vitro* binding assays using *in vitro* translated myc-Bicc1 and recombinant PC1 and PC2 domains. GST-pulldowns confirmed a direct interaction between myc-Bicc1 and GST-CT1 but not GST-CT1-R4227X (Fig. 1d, e). Similarly, myc-Bicc1 interacted directly with GST-NT2. While binding was stronger with the distal sequence (NT2 aa101-223) both N-terminal fragments contributed to the overall binding to Bicc1 (Fig. 1d, e). Interestingly, no direct interaction between Bicc1 and GST-CT2 was detected (Supplementary Fig. S1b) suggesting that the observed *in vivo* interaction with Bicc1 is indirect. Finally, immunoprecipitation using lysates from human kidney epithelial cells (UCL93) to assay endogenous, non-overexpressed proteins showed that PC1, PC2 and BICC1 form protein complexes *in vivo* (Fig. 1f, g).

Different Interaction Motifs for the Binding of Bicc1 to the Polycystins

To define the PC1/PC2 interaction domain(s) in Bicc1, we generated deletion constructs lacking the SAM domain (myc-mBicc1-ΔSAM, aa1-815) or the KH/KHL domains (myc-mBicc1-ΔKH, aa352-977) (Fig. 2a) and studied them by co-IP. Full-length PC1 co-immunoprecipitated with full-length myc-mBicc1 (Fig. 2b,c). Deleting the SAM domain did not significantly reduce the association to PC1 (~55%, p=0.79) compared to full-length myc-mBicc1. However, an 8-fold stronger interaction was observed between full-length PC1 and myc-mBicc1-ΔKH compared to myc-mBicc1 or myc-mBicc1-ΔSAM. These results suggested that the interaction between PC1 and Bicc1 may involve the SAM but not the KH/KHL domains (nor the first 132 amino acids of Bicc1). Potentially the N-terminus (aa1-351) could have an inhibitory effect on PC1-BICC1 association.

Similar experiments were performed to define the Bicc1 interacting domains for PC2 (Fig. 2d,e). Full-length PC2 (PC2-HA) interacted with full-length myc-mBicc1. Unlike PC1, PC2 interacted with myc-mBicc1-ΔSAM, but not myc-mBicc1-ΔKH suggesting that PC2 binding is dependent on the N-terminal domains (aa1-351) but not the SAM domain or distal C-terminus (aa816-977). Co-expression of Bicc1 deletion constructs lacking the SAM domain (myc-mBicc1-ΔSAM) or the KH

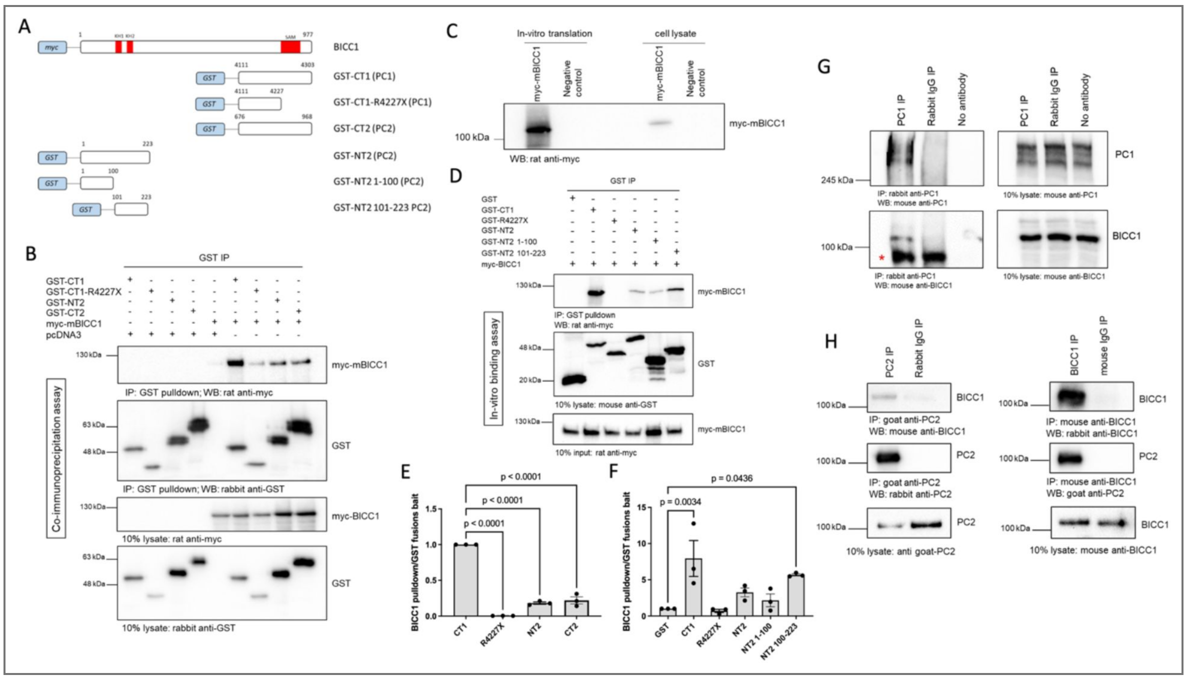

Figure 1. ***Bicc1* Forms a Complex with Polycystin-1 and Polycystin-2.**

Full-length and deletion myc-tagged constructs of Bicc1 were co-expressed with either full-length HA-tagged PC1 or PC2 in HEK-293 cells and tested for their ability to interact by co-IP. (**a**) Schematic diagram of the constructs used in this experiment. (**b**) Western blot analysis following co-IP experiments, using GST tagged constructs as bait, identified protein interactions between PC1 or PC2 domains and *Bicc1*. pcDNA3 was included as a negative control. CT = C-terminus, NT = N-terminus, GST = Glutathione S-Transferase. Co-IP experiments (n=3) were quantified in (c). (**d**) Western blot analysis following *in-vitro* pulldown experiments, using purified GST tagged constructs as bait, identified direct protein interactions between PC1 or PC2 domains and *in vitro* translated myc-Bicc1. In-vitro binding experiments (n=3) were quantified in (e,f). (**g**) Western blot analysis following co-IP experiments, using a rabbit PC1 antibody (2b7) as bait, identified protein interactions between endogenous PC1 and BICC1 in UCL93 cells. A non-immune rabbit IgG antibody or no antibody was included as a negative control; * denotes a non-specific IgG band which is not present in the no antibody control lane. (**h**) Western blot analysis following co-IP experiments, using an anti-mouse Bicc1 or anti-goat PC2 antibody as bait, identified protein interactions between endogenous PC2 and BICC1 in UCL93 cells. Non-immune goat and mouse IgG was included as a negative control.

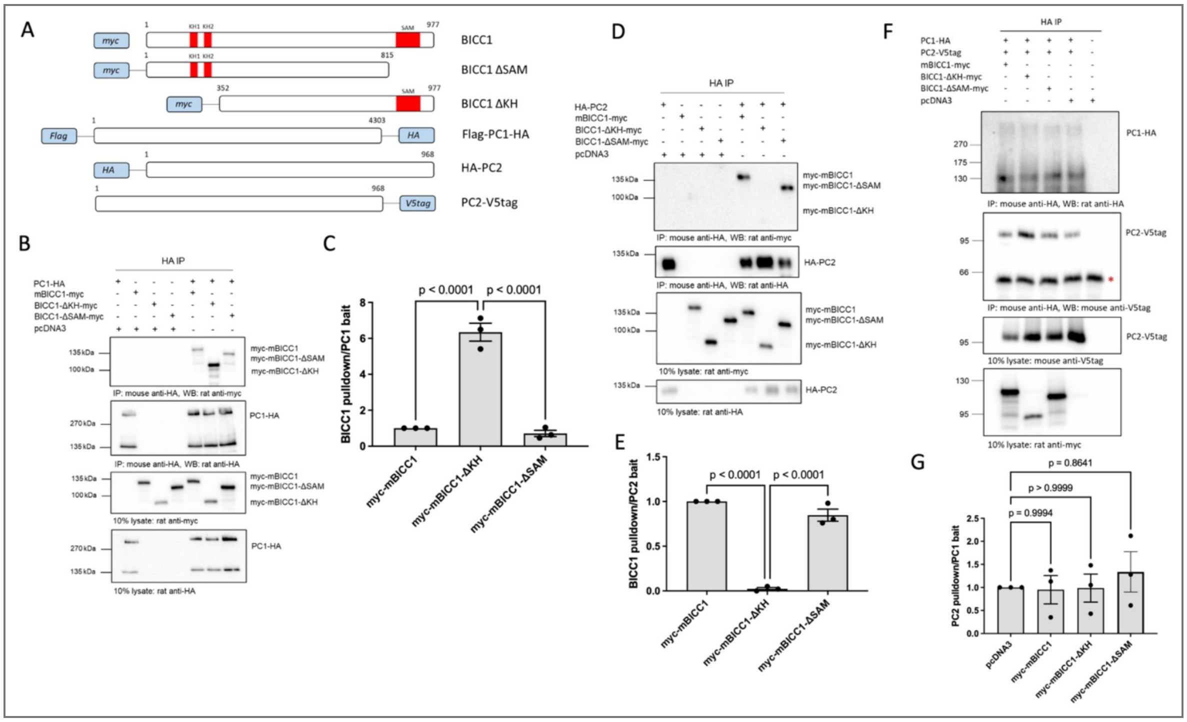

Figure 2. **Interactions between *Bicc1* and Polycystin1/2 Require Different Binding Motifs.**

Full-length and deletion myc-tagged constructs of Bicc1 were co-expressed with either full-length HA-tagged PC1 or PC2 in HEK-293 cells and tested for their ability to interact by co-IP. (**a**) Schematic diagram of the constructs used in this set of experiments with the amino acid positions of full-length Bicc1 or the different deletions indicated. (**b,c**) Western blot analysis following co-IP experiments, using a PC1-HA-tagged construct as bait, identified protein interactions between PC1 and Bicc1 domains. pcDNA3 was included as a negative control (**b**). co-IP experiments (n=3) were quantified in (**c**). (**d,e**) Western blot analysis following co-IP experiments, using a PC2-HA tagged construct as bait, identified protein interactions between PC2 and Bicc1 domains (**d**). pcDNA3 was included as a negative control. Quantification of the co-IP experiments (n=3) is shown in (**e**). (f, g) Western blot analysis following co-IP experiments, using a PC1-HA-tagged construct as bait. The interaction between PC1 and PC2 was not altered in the presence of either full-length Bicc1 or its deletion domains. pcDNA3 was included as a negative control. Asterix represents non-specific interaction with mouse IgG. **(f)**. co-IP experiments (n=3) were quantified in **(g).** One way ANOVA comparisons were performed to assess significance and *P* values are indicated. Error bars represent standard error of the mean.

domains (myc-mBicc1-ΔKH) however had no effect on the interaction of PC1 with PC2 in co-immunoprecipitation assays (Fig 2f,g) suggesting that these interactions are not mutually exclusive.

Cooperativity of BICC1 with other PKD genes

Since our biochemical analysis indicated a direct interaction between BICC1, PC1 and PC2, we wondered whether this is biologically relevant. If this were the case, BICC1 should cooperate with other PKD genes and reducing BICC1 activity in conjunction with reducing either PKD1 or PKD2 activity should still cause a cystic phenotype. We first addressed this question in the *Xenopus* system (Fig. 3), which is an easily manipulatable model to study PKD. The PKD phenotype in frog is characterized by dilated kidney tubules, the loss of the expression of the sodium bicarbonate cotransporter 1 (Nbc1) in the in the distal tubule and the emergence of body-wide edema as a sign of a malfunctioning kidney.[21,22,38,39] Knockdown of Bicc1, Pkd1, Pkd2 or the ARPKD gene Pkhd1 caused a PKD phenotype (Fig. 3e-i" and Supplementary Fig. S2a). The latter, *Pkhd1* was included to assay not only ADPKD, but also ARPKD, which is generally thought to disturb the same cellular mechanisms. To test whether Bicc1 cooperated with the PKD genes we then performed combined knockdowns. We titrated each of the four MOs to a concentration that on its own resulted in little phenotypic changes upon injection into *Xenopus* embryos (Fig. 3j,k and Supplementary Fig. S2b). However, combining *Bicc1-MO1+2* with *Pkd1-sMO*, *Pkd2-MO* or *Pkhd1-sMO* at suboptimal concentrations resulted in the re-emergence of a strong PKD phenotype.

While injections with individual MOs developed edema in about 10% of the embryos, co-injections caused edema formation in almost 100% of the embryos (Fig. 3j, last 3 columns). A similar result was seen for the expression of *Nbc1* in the late distal tubule, where individual MO injections showed some changes in gene expression, but double MO injections had a highly synergistic effect resulting in a near complete loss of *Nbc1* (Fig. 3k). Of note, this approach is based on a sensitized biological readout, but not on reducing expression levels to a fixed amount (e.g., 50%).

We next investigated whether a similar cooperation between Bicc1 and Pkd1 or Pkd2 can be observed in genetic mouse models. We initially focused on Bicc1 and Pkd2. Both *Bicc1* and *Pkd2* knockout mice develop cystic kidneys as early as E15.5.[22,40] As this is the earliest time point cystic kidneys can be observed, crossing those strains did not allow us to assess cooperativity (data not shown). Moreover, like in the case of compound *Pkd1/Pkd2* mutants,[41] kidneys from *Bicc1*$^{+/-}$*:Pkd2*$^{+/-}$ did not exhibit cysts (data not shown). Thus, we instead used mice carrying the Bicc1 hypomorphic allele *Bpk*, which develop a cystic kidney phenotype postnatally.[18,42] To assess cooperativity, we removed one copy of *Pkd2* in the *Bpk* mice. Comparing the kidneys of *Bicc1*$^{Bpk/Bpk}$*:Pkd2*$^{+/-}$ to those of *Bicc1*$^{Bpk/Bpk}$*:Pkd2*$^{+/+}$ at postnatal day P14 revealed that the compound mutant kidneys were larger and more translucent (Fig. 4a) and the kidney/body weight ratios (KW/BW) were significantly increased (Fig. 4b). Moreover, analyzing survival the compound mutants showed a trend towards an earlier demise (Supplementary Table S1). We did not detect sex differences in the phenotype (Supplementary Figure S3c). Yet, the reduction in *Pkd2* gene dose affected the progression of the disease, but not its onset. Performing the same analysis at postnatal day P4 did not show any differences (Fig. 4c).

Next, we performed a similar mouse study for *Pkd1* using the *Pkd1*$^{Fl/Fl}$*:Pkhd1-Cre* line described previously[43] (in the following referred to as *Pkd1*$^{CD-}$). This mouse line eliminates *Pkd1* postnatally in the collecting ducts. Similar to the *Bicc1/Pkd2* scenario, when removing one copy of *Pkd1* in the collecting ducts, the *Bicc1*$^{Bpk/Bpk}$*:Pkd1*$^{+/CD-}$ appeared larger when comparing kidneys from littermates (Fig. 4d) and littermates exhibited statistically significant differences in KW/BW ratio (Fig. 4e). Yet, the phenotype was rather subtle and aggregating all the data did not show differences in KW/BW ratios between *Bicc1*$^{Bpk/Bpk}$*:Pkd1*$^{+/+}$ and *Bicc1*$^{Bpk/Bpk}$*:Pkd1*$^{+/CD-}$ mice (Supplementary Fig. S3d). Thus, to further corroborate the genetic interaction, we determined the cystic index for proximal tubules and collecting ducts using LTA and DBA staining, respectively. This showed an increase in collecting duct cysts upon removal of one copy of *Pkd1* (Fig. 4g). Like in the case of *Pkd2*, the phenotype seems to be correlated with cyst expansion and not the onset, as there was no difference at postnatal day P7 (Fig. 4f) and we did not detect

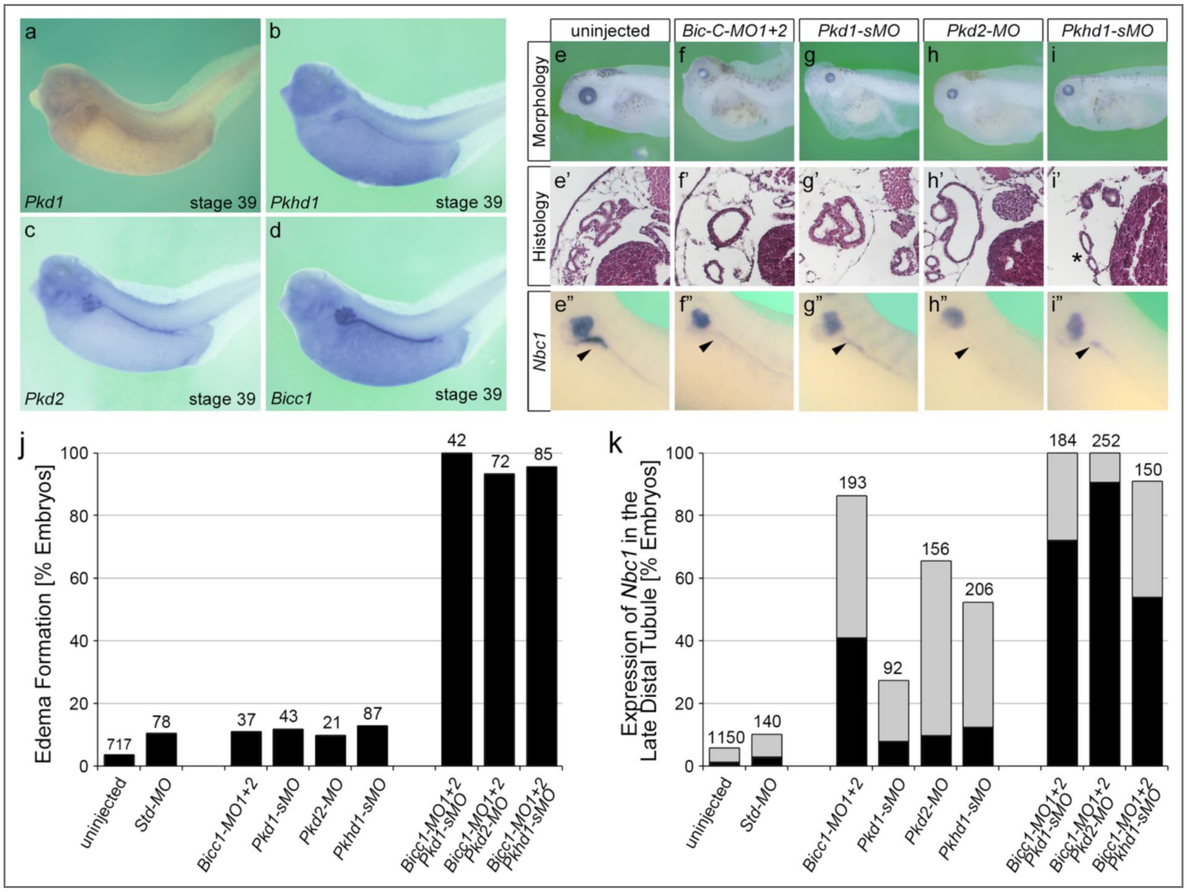

Figure 3. **Cooperativity of Bicc1 and PKD Genes in *Xenopus*.**

(**a-d**) mRNA expression of *Pkd1*, *Pkhd1, Pkd2* and *Bicc1* in the *Xenopus* pronephros at stage 39. (**e-i"**) Knockdown of Bicc1 (**f-f"**), Pkd1 (**g-g"**), Pkd2 (**h-h"**) and Pkhd1 (**i-i"**) by antisense morpholino oligomers (MOs) results in a PKD phenotype compared to uninjected control *Xenopus* embryos (**e-e"**). The phenotype is characterized by the formation of edema due to kidney dysfunction (**e,f,g,h,i**; stage 43), the development of dilated renal tubules (**e',f',g',h',i'**; stage 43) and the loss of *Nbc1* in the late distal tubule by whole mount *in situ* hybridizations (arrowheads in **e",f",g",h",i"**; stage 39). **(j,k)** To examine cooperativity, *Xenopus* embryos were injected with suboptimal amounts of the MOs, either alone or in combination, and analyzed for edema formation at stage 43 (**j**) and the expression of *Nbc1* at stage 39 (**k**) with gray bars showing reduced and black bars showing absent *Nbc1* expression in the late distal tubule. Data are the accumulation of multiple independent fertilizations with the number of embryos analyzed indicated above each condition.

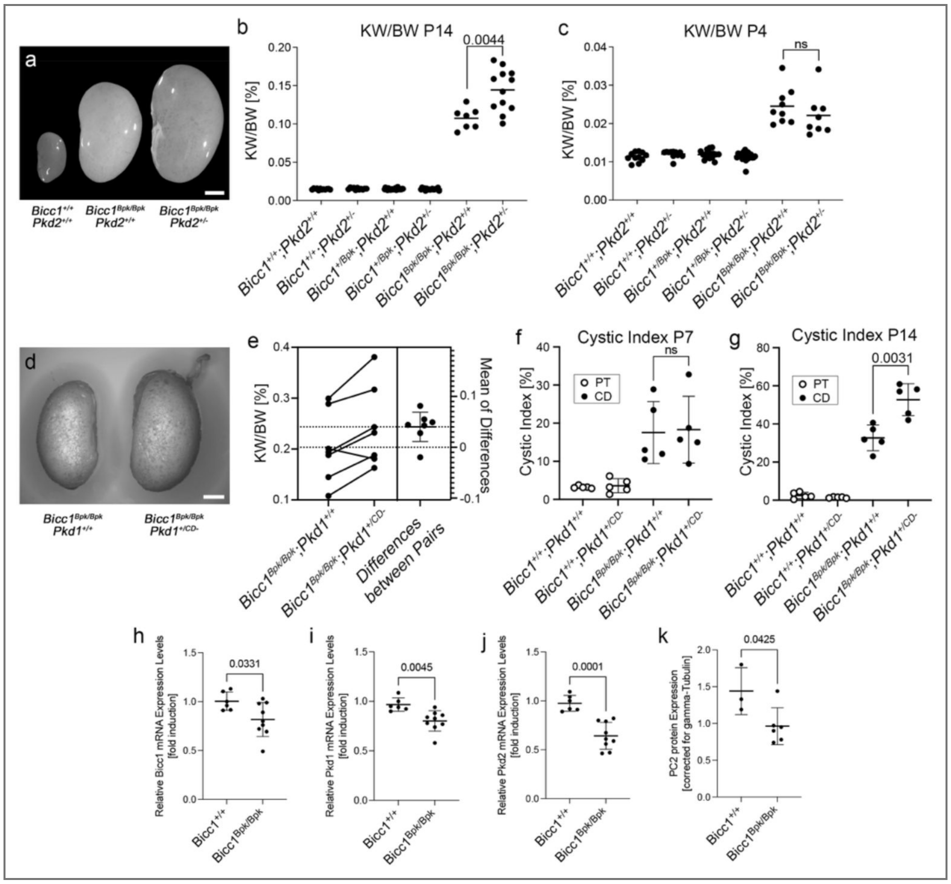

Figure 4. **Cooperativity of Bicc1 and Pkd1 and Pkd2 in Mouse.**

(**a-c**) Bicc1 and Pkd2 interact genetically. Offspring from *Bicc1;Pkd2* compound mice at postnatal day P4 and P14 are compared by outside kidney morphology at postnatal day P14 (**a,** scale bar is 2 mm), and kidney to body weight ratio (KW/BW) at P14 (**b**) and P4 (**c**). (**d-g**) Bicc1 and Pkd1 interact genetically. *Bicc1;Pkd1* compound mice are compared by outside kidney morphology at P14 showing a kidney from *Bicc1$^{Bpk/Bpk}$:Pkd1$^{+/+}$*and a *Bicc1$^{Bpk/Bpk}$:Pkd1$^{+/CD-}$*littermates (**d,** scale bar is 2 mm, no kidney of a wildtype littermate is shown, as it was not present in the litter), estimation plot of KW/BW ratio comparing littermates at P14 with a *P*-value = 0.092 (**e**), and cystic index, i.e. percent of of proximal tubules (PT) and collecting ducts (CD) cysts in respect to the total kidney area at P7 (**f**) and P14 (**g**). Two-sided paired t-tests were performed to assess significance, and the *P*-values are indicated; error bars represent standard deviation. (**h-k**) qRT-PCR analysis for *Bicc1*, *Pkd1*, and *Pkd2* expression (**h-j**) and quantification of the PC2 expression levels by Western blot (**k**) in kidneys at P4 before the onset of a strong cystic kidney phenotype. Data were analyzed by t-test and the *P*-values are indicated. Please note that the y-axes of the different panels are intentionally different to best visualize the changes between the groups analyzed.

increased mortality in the compound mutants (Supplementary Table S2). It is noteworthy that neither the *Bicc1/Pkd2* nor the *Bicc1/Pkd1* compound mutants showed an aggravated kidney function based on Blood Urea Nitrogen (BUN) levels (Supplementary Fig. S3a,b,e) likely due to the aggressive nature of the *Bicc1*$^{Bpk/Bpk}$ phenotype. Of note, due to the different genetic approaches using a *Pkd2* null allele and a conditional *Pkd1* allele, the outcomes of the two crosses cannot be directly compared. Yet, these *in vivo* data support our biochemical interaction data and demonstrated that *Bicc1* cooperates with *Pkd1* and *Pkd2*.

Finally, to better understand how Bicc1 would exert such a phenotype, we analyzed the expression of the PKD genes in the *Bicc1*$^{Bpk/Bpk}$mice. We have previously demonstrated that *Pkd2* levels are reduced in a complete Bicc1 null mice.[22] Performing qRT-PCR of kidneys from wildtype and *Bicc1*$^{Bpk/Bpk}$ at P4 (i.e. before the onset of a strong cystic phenotype), revealed that *Bicc1*, *Pkd1* and *Pkd2* were statistically significantly down-regulated (Fig. 4h-j). The effect on *Pkd2* mRNA was confirmed by protein analysis for PC2 (Fig. 4k and Supplementary Fig. S3f).

BICC1 Variants in Patients with early and severe Polycystic Kidney Disease

To evaluate whether these interactions are relevant for human PKD, we analyzed an international cohort of 2,914 PKD patients by massive parallel sequencing (MPS) [44,45] focusing on VEO-ADPKD patients with the hypothesis that *BICC1* variants may lead to a more severe and earlier PKD phenotype. While variants in *BICC1* are very rare, we could identify two patients with *BICC1* variants harboring an additional *PKD2* or *PKD1* variant *in trans*, respectively. None of these *BICC1* variants were detected in the control cohort or in non-VEO-ADPKD patients. Moreover, besides the variants reported below, the patients had no other variants in any of other PKD genes or genes which phenocopy PKD including *PKD1*, *PKD2*, *PKHD1*, *HNF1ß*, *GANAB*, *IFT140*, *DZIP1L*, *CYS1*, *DNAJB11*, *ALG5*, *ALG8*, *ALG9*, *LRP5*, *NEK8*, *OFD1* or *PMM2*.

The first patient was severely and prenatally affected demonstrating a Potter sequence with huge echogenic kidneys and oligo-/anhydramnios. Autopsy confirmed VEO-ADPKD with absence of ductal plate malformation invariably seen in ARPKD. The fetus carried the *BICC1* variant (c.2462G>A, p.Gly821Glu) inherited from his father, who presented with two small renal cysts in one of his kidneys, and a *PKD2* variant (c.1894T>C, p.Cys632Arg) that arose *de novo* (Fig. 5a). Individual *in silico* predictions (SIFT, Polyphen2, CADD, Eigen-PC, FATHMM, GERP++ RS and EVE), meta scores (REVEL, MetaSVM, and MetaLR) and other protein function predictions (PrimateAI, AlphaMissense, ESM1b and ProtVar) indicate that this *PKD2* missense variant is likely pathogenic (Supplementary Table S3). Moreover, structural analysis suggests that the hydrophilic substitution may interfere with the Helix S5 pore domain of PKD2 and change its ion channel function (Fig. 5b,c). Finally, *PKD2* p.Cys632Arg has been previously reported as part of a PKD2 pedigree and implicated as a critical determinant for Polycystin-2 function.[46,47] On the other hand, the *BICC1* p.Gly821Glu variant is located in an intrinsically disordered domain of BICC1 between the KH and the SAM domains (Fig. 6f). To address whether the variant is hypomorphic, we used CRISPR-Cas9-mediated gene editing to generate HEK293T cells lacking BICC1 or harboring the p.Gly821Glu mutation (BICC1-G821E). These cells were analyzed for their impact on the translation of *PKD2*, a well-established target of Bicc1.[22] As shown in Fig. 5d,e PC2 protein levels were strongly reduced in two independent HEK293T BICC1-G821E cells when compared to unedited HEK293T cells. Most notably, the PC2 levels were comparable to the levels found in HEK293T carrying a *BICC1* null allele (HEK293T BICC1-KO) (Supplementary Figure S2c,d). Based on these data we hypothesize that the major disease effect results from the pathogenic *PKD2* variant but is aggravated by the *BICC1* variant.

The second patient presented perinatally with massively enlarged hyperechogenic kidneys, while the parents, both in their thirties, and the remaining family members were reported to be healthy (Fig. 5f-h). He carried a paternal canonic *BICC1* pathogenic splicing variant *(*c.1179+1G>T), which is likely pathogenic as the protein is truncated after exon 10, and a novel heterozygous *PKD1* variant (c.11942C>T, p.Ala3981Val) which has not been previously reported (Fig. 5f). While the *PKD1* variant appears minor in its amino acid change (i.e. Ala to Val), *in silico* analyses using

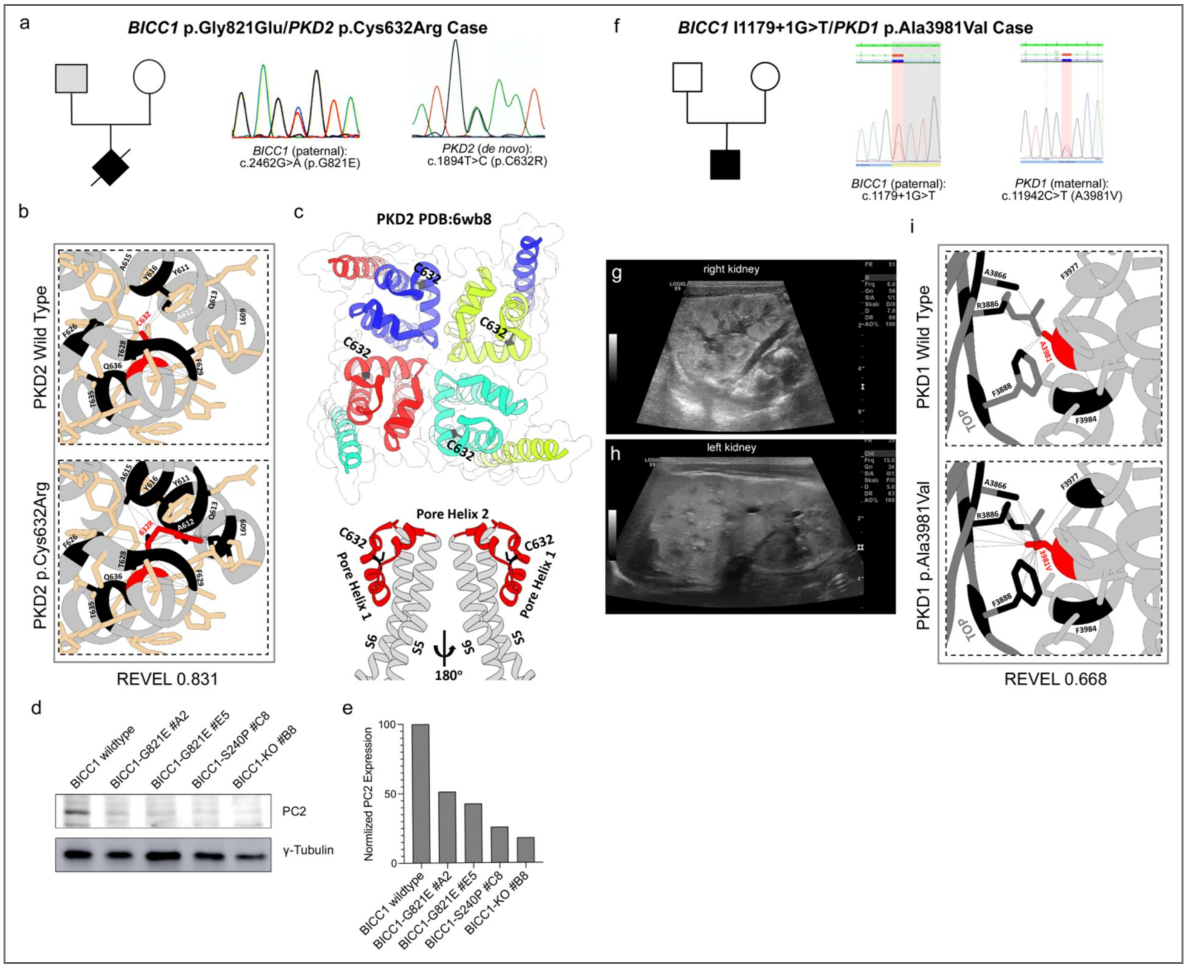

Figure 5. Identification of Human *BICC1* Variants.

(**a-c**) *BICC1* p.G821E/*PKD2* p.C632R case with pedigree and the electropherograms (**a**), the structural analysis of the PKD2 showing the local structure around the cysteine at position 632 (indicated in red) and its putative impact in the variant including the REVEL score (**b**) as well as its location within the global PC2 structure highlighting the potential of the variant impacting the PC2 ion channel function (**c**). (**d,e**) Western blot analysis for PC2 comparing wildtype HEK293T, HEK293T BICC1 p.Gly821Glu (BICC1-G821E), HEK293T BICC1 p.Ser240Pro (BICC1-S240P) and HEK293T BICC1 knockout (BICC1-KO) cells and quantification thereof. □-Tubulin was used as loading control. (**f-i**) *BICC1* I1179+1G>T/*PKD1* p.Ala3981Val case with pedigree and the electropherograms (**f**), the ultrasound analysis of the left and right kidneys (**g,h**) and the structural analysis of the PC1 showing the local structure around the alanine at position 3981 (indicated in red) and its putative impact in the variant including the REVEL score (**i**).

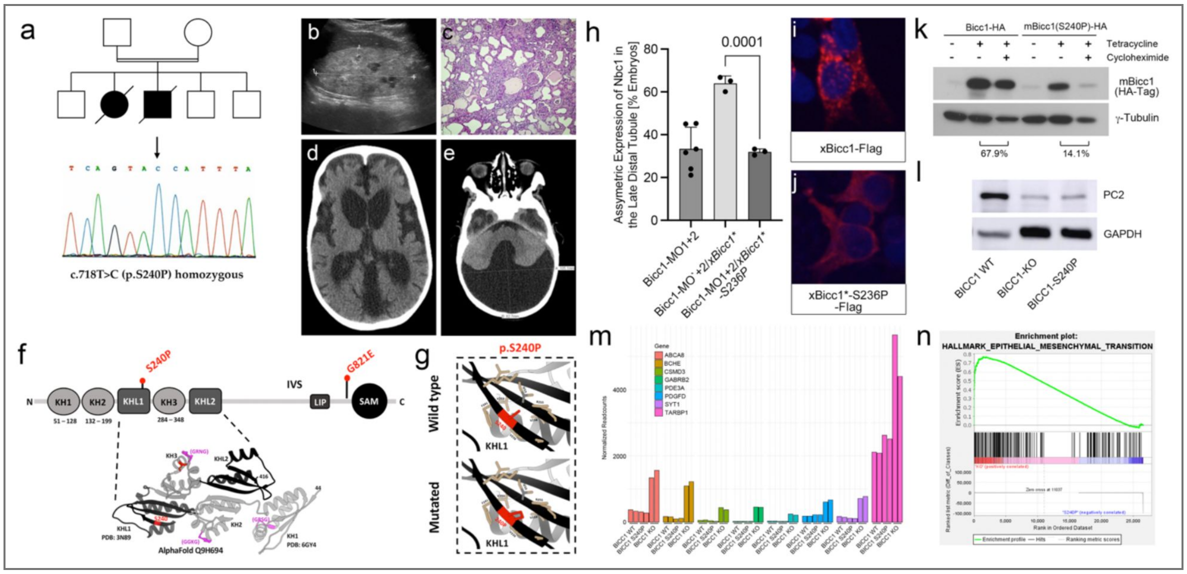

Figure 6. The Homozygous *BICC1* p.Ser240Pro Variant is a Hypomorphic Cystic Disease-Causing Variant.

(**a-e**) Consanguineous multiplex pedigree with two siblings affected by VEO-ADPKD identified the homozygous *BICC1* missense variant c.718T>C (*BICC1* p.Ser240Pro) absent from gnomAD and other internal and public databases. Electropherogram is shown in (**a**). The affected girl presented at a few months of age with renal failure and enlarged polycystic kidneys that lacked corticomedullary differentiation (**c**). Histology after bilateral nephrectomy showed polycystic kidneys more suggestive of ADPKD than ARPKD without any dysplastic element. Her younger brother exhibited enlarged hyperechogenic polycystic kidneys prenatally by ultrasound (**b**). In addition, in his early infancy arterial hypertension and a Dandy-Walker malformation with a low-pressure communicating hydrocephalus were noted (**d,e**). (**f**) Ribbon diagram and schematic diagram of BICC1 showing the KH, KHL and SAM domains. The two *BICC1* variants identified in this study, BICC1 p.Ser240Pro (S240P) and BICC1 p.Gly821Glu (G821E) are indicated in red. (**g**) Solid boxes correspond to local impacts of p.Ser240Pro (p.S240P) on BICC1 structure, interactions are labeled as dashed lines (pseudobonds). GXXG motifs colored in magenta, representative missense variant residues colored in red and residues adjacent to selected variant (<5Å) colored in tan. (**h**) Rescue experiments of *Xenopus* embryos lacking BicC1 by co-injections with the wild type or mutant constructs. Embryos were scored for the re-expression of *Nbc1* in the late distal tubule by whole mount *in situ* hybridizations. Quantification of at least 3 independent experiments is shown. **(i,j**) HEK293T cells were transfected with Flag-tagged constructs of wild type or mutant Bicc1 and the subcellular localization of Bicc1 was visualized (red). Nuclei were counterstained with DAPI (blue). (**k**) Protein stability analysis using tetracycline-inducible HEK293T cells comparing the expression levels of Bicc1 and Bicc1-S240P 24 hours after removal of tetracycline and addition of cycloheximide. □-Tubulin was used as loading control. The percentage of protein destabilization because of protein synthesis inhibition by cycloheximide is indicated. (**l**) Western Blot analysis of wildtype HEK293T, cells lacking BICC1 (BICC1-KO) and isogenic cells with the BICC1 p.Ser240Pro (BICC1-S240P) variant for PC2 expression. GAPDH was used as loading control. (**m,n**) Bar graph of the mRNA-seq transcriptomic analysis comparing BICC1 wildtype, knockout and S240P isogenic HEK293T cells showing the eight most significantly upregulated transcripts (based on their Padj levels) in the BICC1 KO cells (**m**). For each gene, the normalized expression levels from each of the 6 samples (2 wildtype, KO and 240P each) is shown. (**n**) GSEA plot showing the enrichment of the Hallmark Epithelial_Mesenchymal_Transition data set in the BICC1-KO cells *vs.* the BICC1-S240P cells.

individual predictions (SIFT, Polyphen2, CADD and EVE), Meta scores (REVEL) and other protein function predictions (PrimateAI, Alphamissense and ESM1b) indicate that the missense variant is likely pathogenic (Supplementary Table S3). Structural analyses suggest that although the Ala3981Val variant does not destabilize the Helix structure, its contact with the TOP domain could interfere with domain flexibility and PC1 complex assembly.

A Sibling Pair of PKD Patients with a Homozygous *BICC1* Variant

The most insightful finding for a critical role for BICC1 in human PKD was the discovery of a homozygous *BICC1* variant in a consanguinous Arab multiplex pedigree, two siblings, a boy and a girl, diagnosed with VEO-ADPKD (Fig. 6a-e). The affected female presented at a few months of age with kidney failure and enlarged polycystic kidneys that lacked corticomedullary differentiation. Histology after bilateral nephrectomy showed polycystic kidneys more suggestive of ADPKD than ARPKD without any dysplastic element (Fig. 6c). Her younger brother exhibited enlarged hyperechogenic polycystic kidneys antenatally by ultrasound (Fig. 6b). In addition, during early infancy, arterial hypertension and a Dandy-Walker malformation with a low-pressure communicating hydrocephalus were noted (Fig. 6d,e). By customized MPS, we identified the homozygous missense *BICC1* variant (c.718T>C, p.Ser240Pro) (Fig. 6a). This variant was absent from gnomAD and fully segregated with the cystic phenotype present in this family. It results in a non-conservative change from the aliphatic, polar-hydrophilic serine to the cyclic, apolar-hydrophobic proline located in the second beta sheet of the first KHL1 domain and very likely disrupts the beta sheet and thus the RNA-binding activity of Bicc1 (Fig. 6f,g and Supplementary Table S4). In the more severely affected younger brother, we also detected an additional heterozygous *PKD2* variant (c.1445T>G, p.Phe482Cys), which results in a non-conservative change from phenylalanine to cysteine (Supplementary Table S3). It was previously reported that this PC2 Phe482Cys variant exhibited altered kinetic PC2 channel properties, increased expression in IMCD cells and a different subcellular distribution when compared to wild-type PC2,[48] These features suggested altered properties of this PC2 variant, yet its contribution to the case reported here remain untested.

Unfortunately, both siblings passed away and besides DNA and the phenotypic analysis described above neither human tissue nor primary patient-derived cells could be collected. Thus, to validate the pathogenicity of this point mutation, we turned to the amphibian model of PKD.[21,22] In *Xenopus*, knockdown of Bicc1 using antisense morpholino oligomers (*Bicc1-MO1+2*) causes a PKD phenotype, which can be rescued by co-injection of synthetic mRNA encoding *Bicc1*.[21] To test whether *BICC1* p.Ser240Pro had lost its biological activity, we introduced the same mutation into the *Xenopus* gene where the Ser is located at position 236 of the *Xenopus* gene (in the following referred to as *xBicC1*-S236P*). *Xenopus* embryos were injected with *Bicc1-MO1+2* at the 2-4 cell stage followed by a single injection of 2 ng wild type or *xBicc1*-S236P* mRNAs at the 8-cell stage. At stage 39 (when kidney development has been completed) embryos were analyzed by whole mount *in situ* hybridization for the expression of *Nbc1* in the late distal tubule of the pronephric kidney, one of the most reliable readouts for the amphibian PKD phenotype.[21] As shown in Fig. 6h, wild type *Bicc1* mRNA restored expression of *Nbc1* on the injected side in 63% of the embryos. However, *xBicc1*-S236P* did not have any effect, and the embryos were indistinguishable from those injected with the *Bicc1-MO1+2* alone. This suggested that *xBicc1*-S236P* was functionally impaired. To address this hypothesis, we first assessed the subcellular localization of Bicc1 to foci that are thought to be involved in mRNA processing.[19,22,30,49] Transfection of Flag-tagged Bicc1 (*xBicc1*-S236P-Flag*) into HEK293T cells reproduced this pattern (Fig. 6i). Surprisingly, xBicc1*-S236P-Flag was no longer detected in these cytoplasmic foci but rather homogenously dispersed throughout the cytoplasm (Fig. 6j). Western blot analysis demonstrated that this was accompanied by a reduction in protein levels (Fig. 6k). *In vitro* transcription/translation detected no differences between the proteins suggesting that the wildtype and Bicc1 S236P-Flag are translated equivalently (data not shown). Yet, in an *in vivo* pulse-chase experiments a mBICC1 p.Ser240Pro variant was less stable than its wildtype counterpart (Fig. 6k). However, whether the reduced protein level was due to an inherent instability of the mutant protein or a consequence of its mislocalization remains to be resolved. Finally, as in the case of BICC1 p.Gly821Glu we engineered HEK293T cells to harbor the *BICC1* p.Ser240Pro variant (BICC1-S240P). Western blot analysis demonstrated a reduction in PC2 levels in the BICC1-S240P cells when compared to unedited cells and that this reduction was comparable to PC2 levels in BICC1-KO cells (Figs. 5d,e and **6l**).

Finally, to determine to what extent the *BICC1* p.Ser240Pro variant differs from a *BICC1* loss of function allele, we performed mRNA sequencing (mRNA-seq) of the genetically engineered HEK293T cells. Differential gene expression analysis identified several genes that were differentially up- or down-regulated in the BICC1-S240P and the BICC1-KO cells compared to their unedited counterpart (Supplementary Fig. 4a,e). Approximately 24% and 18% of the differentially expressed genes were shared between BICC1-S240P or the BICC1-KO cells, respectively (Supplementary Fig. S4b,f). Yet, a substantial number of genes were specific to either cell line. The BICC1-S240P-enriched/depleted transcripts were generally also enriched/depleted in the BICC1-KO cells but did not reach statistical significance (Supplementary Fig. 4c,g). Conversely, many of the BICC1-KO enriched transcripts were specifically enriched/depleted in the BICC1-KO cells and not in the BICC1-S240P cells (Fig. 6m and Supplementary Fig. 4d). This suggested that there are qualitative differences between a null phenotype and the *BICC1* p.Ser240Pro variant, supporting our hypothesis that *BICC1* p.Ser240Pro acts as a hypomorph. Indeed, Gene Set Enrichment Analysis (GSEA) using the hallmark gene sets and comparing BICC1-KO and BICC1-S240P cells revealed a statistically significant enrichment for the Hallmark_Epithelial_Mesenchymal_Transition set (Fig. 6n and Supplementary Table S5), a pathway previously implicated in ADPKD.[50,51]

Discussion

BICC1 has been extensively studied in multiple animal models, which have suggested a critical role for BICC1 in several different developmental processes and in tissue homeostasis.[26] This study functionally implicates it to human disease in general and PKD in particular by identifying the homozygous *BICC1* p.Ser240Pro variant, which was sufficient to cause a cystic phenotype in a sibling pair of human PKD patients. It is noteworthy that another study identified heterozygous *BICC1* variants in two patients with mildly cystic dysplastic kidneys.[23] Yet, both variants were also present in one of the unaffected parents. While such a situation is extremely rare and does not significantly contribute to the mutational load in ADPKD or ARPKD, it demonstrated that loss of BICC1 is sufficient to cause PKD in humans. In addition, variants in *BICC1* and *PKD1 and PKD2* co-segregated in PKD patients from an International Clinical Diagnostic Cohort. While we have not yet shown the impact of each variant when introduced in a compound heterozygous situation, we postulate that PKD alleles *in trans* and/or *de novo* exert an aggravating effect and contribute to polycystic kidney disease. A reduced dosage of PKD proteins would severely disturb the homeostasis and network integrity, and by this correlates with disease severity in PKD. ADPKD is quite heterogeneous and - even within the same family - shows quite some phenotypic variation.[52,53] It is thought that stochastic inputs, environmental factors and genetics influence PKD.[53] The demonstrated interaction of BICC1, PC1 and PC2 now provides a molecular mechanism that can explain some of the phenotypic variability in these families. Of note, while our mouse studies support cooperation between Bicc1, Pkd1 and Pkd2, genetic proof for Bicc1 acting as a disease modifier, i.e. reduction of Bicc1 activity in a homozygous *Pkd1* or *Pkd2* background remains outstanding.

The second important aspect of this study is that BICC1 emerges as a central in the regulation of PKD1/PKD2 activity. Functional studies reported here and previously[22,35,36] demonstrate that Bicc1 regulates the expression of *Pkd1* and *Pkd2*. Moreover, we now show that Bicc1 and PC1/PC2 physically interact and that lowering the expression levels of both proteins is sufficient to cause a PKD phenotype in frogs. Finally, the reduction of the gene dose for Pkd1 or Pkd2 in a hypomorphic mouse allele of *Bicc1* results in a more severe cystic kidney phenotype. These results in the kidney are paralleled and augmented in studies of left/right patterning, where Pkd2 can activate Bicc1 and where Bicc1 triggers critical aspects in establishing laterality.[19,30,54,55] Thus, it is tempting to speculate that BICC1/PKD1/PKD2 are components of a critical regulatory network in maintaining epithelial homeostasis.

BICC1 has emerged as an important posttranscriptional regulator modifying gene expression through modulating the effects of microRNAs (miRNAs), regulating mRNA polyadenylation and translational repression and activation.[22,26,32,56–59]. While *PKD2* is the most appealing target in respect to ADPKD,[22] there are undoubtable others (e.g. adenylate cyclase-6)[32] that may be equally critical. Lastly, Bicc1 has been implicated in the regulation of miRNAs such as those of the *miR-17* family.[22] This is of particular interest as a connection between *miR-17* activity and PKD is well-established.[60–66] Both *Pkd1* and *Pkd2* mRNA are targeted by *miR-17*,[67] and an *anti-miR-17* oligonucleotide is being developed as a PKD therapeutic.[68] While we have shown that Bicc1 and *miR-17* targets *Pkd2* mRNA,[22] a similar scenario for *Pkd1* is possible, but not yet shown. Thus, a tempting hypothesis is that the interaction between Bicc1, Pkd1 and Pkd2 and miRNAs - even though not examined in this study – compartmentalizes Bicc1's activity where Bicc1 is posttranscriptionally inactive when complexed to Pkd1/Pkd2, but modulates Pkd1/Pkd2 expression when unbound. Such a regulatory complex could be responsible for several of the aspects of human ADPKD. In the future, it would be interesting to see how Bicc1 and its posttranscriptional targets are integrated and together contribute towards preventing kidney epithelial cells from developing a cystic phenotype.

Data availability statement

The datasets are presented in the figures and the supplementary information. The mRNA-seq data are deposited into the Gene Expression Omnibus (GEO) database (GSE262417) and are available online. Additional information is available from the corresponding author on reasonable request.

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Supplemental material

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References

1. Harris PC, Torres VE (2009) Polycystic kidney disease. *Annu Rev Med* **60**:321-337 https://doi.org/10.1146/annurev.med.60.101707.125712
2. Ong AC, Devuyst O, Knebelmann B, Walz G, Diseases E-EWGfIK (2015) Autosomal dominant polycystic kidney disease: the changing face of clinical management. *Lancet* **385**:1993-2002 https://doi.org/10.1016/S0140-6736(15)60907-2
3. Torres VE, Harris PC, Pirson Y (2007) Autosomal dominant polycystic kidney disease. *Lancet* **369**:1287-1301 https://doi.org/10.1016/S0140-6736(07)60601-1
4. Bergmann C, Guay-Woodford LM, Harris PC, Horie S, Peters DJM, Torres VE. (2018) Polycystic kidney disease. *Nat Rev Dis Primers* **4**:50 https://doi.org/10.1038/s41572-018-0047-y
5. Bergmann C, Zerres K (2007) Early manifestations of polycystic kidney disease. *Lancet* **369** https://doi.org/10.1016/S0140-6736(07)61005-8
6. Ogborn MR (1994) Polycystic kidney disease--a truly pediatric problem. *Pediatr Nephrol* **8**:762-767 https://doi.org/10.1007/BF00869116
7. Rossetti S, Kubly VJ, Consugar MB, et al. (2009) Incompletely penetrant PKD1 alleles suggest a role for gene dosage in cyst initiation in polycystic kidney disease. *Kidney Int* **75**:848-855 https://doi.org/10.1038/ki.2008.686
8. Vujic M, Heyer CM, Ars E, et al. (2010) Incompletely penetrant PKD1 alleles mimic the renal manifestations of ARPKD. *J Am Soc Nephrol* **21**:1097-1102 https://doi.org/10.1681/ASN.2009101070
9. Bergmann C (2015) ARPKD and early manifestations of ADPKD: the original polycystic kidney disease and phenocopies. *Pediatr Nephrol* **30**:15-30 https://doi.org/10.1007/s00467-013-2706-2
10. Durkie M, Chong J, Valluru MK, Harris PC, Ong ACM (2021) Biallelic inheritance of hypomorphic PKD1 variants is highly prevalent in very early onset polycystic kidney disease. *Genet Med* **23**:689-697 https://doi.org/10.1038/s41436-020-01026-4
11. Ong AC, Harris PC (2015) A polycystin-centric view of cyst formation and disease: the polycystins revisited. *Kidney Int* **88**:699-710 https://doi.org/10.1038/ki.2015.207
12. Senum SR, Li YSM, Benson KA, et al. (2022) Monoallelic IFT140 pathogenic variants are an important cause of the autosomal dominant polycystic kidney-spectrum phenotype. *Am J Hum Genet* **109**:136-156 https://doi.org/10.1016/j.ajhg.2021.11.016

13. **Besse W,** Chang AR, Luo JZ, et al. (2019) ALG9 Mutation Carriers Develop Kidney and Liver Cysts. *J Am Soc Nephrol* **30**:2091-2102 https://doi.org/10.1681/ASN.2019030298

14. **Gall E Cornec-Le,** Olson RJ, Besse W, et al. (2018) Monoallelic Mutations to DNAJB11 Cause Atypical Autosomal-Dominant Polycystic Kidney Disease. *Am J Hum Genet* **102**:832-844 https://doi.org/10.1016/j.ajhg.2018.03.013

15. **Porath B,** Gainullin VG, Cornec-Le Gall E, et al. (2016) Mutations in GANAB, Encoding the Glucosidase IIalpha Subunit, Cause Autosomal-Dominant Polycystic Kidney and Liver Disease. *Am J Hum Genet* **98**:1193-1207 https://doi.org/10.1016/j.ajhg.2016.05.004

16. **Nauta J,** Ozawa Y, Sweeney WE, Rutledge JC, Avner ED. (1993) Renal and biliary abnormalities in a new murine model of autosomal recessive polycystic kidney disease. *Pediatric nephrology* **7**:163-172

17. **Flaherty L,** Bryda EC, Collins D, Rudofsky U, Montogomery JC (1995) New mouse model for polycystic kidney disease with both recessive and dominant gene effects. *Kidney Int* **47**:552-558

18. **Cogswell C,** Price SJ, Hou X, Guay-Woodford LM, Flaherty L, Bryda EC (2003) Positional cloning of jcpk/bpk locus of the mouse. *Mamm Genome* **14**:242-249

19. **Maisonneuve C,** Guilleret I, Vick P, et al. (2009) Bicaudal C, a novel regulator of Dvl signaling abutting RNA-processing bodies, controls cilia orientation and leftward flow. *Development* **136**:3019-3030 https://doi.org/10.1242/dev.038174

20. **Bouvrette DJ,** Sittaramane V, Heidel JR, Chandrasekhar A, Bryda EC (2010) Knockdown of bicaudal C in zebrafish (Danio rerio) causes cystic kidneys: a nonmammalian model of polycystic kidney disease. *Comparative medicine* **60**:96-106

21. **Tran U,** Pickney LM, Ozpolat BD, Wessely O (2007) Xenopus Bicaudal-C is required for the differentiation of the amphibian pronephros. *Dev Biol* **307**:152-164 https://doi.org/10.1016/j.ydbio.2007.04.030

22. **Tran U,** Zakin L, Schweickert A, et al. (2010) The RNA-binding protein bicaudal C regulates polycystin 2 in the kidney by antagonizing miR-17 activity. *Development* **137**:1107-1116 https://doi.org/10.1242/dev.046045

23. **Kraus MR,** Clauin S, Pfister Y, et al. (2012) Two mutations in human BICC1 resulting in Wnt pathway hyperactivity associated with cystic renal dysplasia. *Hum Mutat* **33**:86-90 https://doi.org/10.1002/humu.21610

24. **Fu Y,** Kim I, Lian P, et al. (2010) Loss of Bicc1 impairs tubulomorphogenesis of cultured IMCD cells by disrupting E-cadherin-based cell-cell adhesion. *Eur J Cell Biol* **89**:428-436 https://doi.org/10.1016/j.ejcb.2010.01.002

25. **Gamberi C,** Hipfner DR, Trudel M, Lubell WD (2017) Bicaudal C mutation causes myc and TOR pathway up-regulation and polycystic kidney disease-like phenotypes in Drosophila. *PLoS Genet* **13**:e1006694 https://doi.org/10.1371/journal.pgen.1006694

26. **Dowdle ME,** Kanzler CR, Harder CRK, Moffet S, Walker MN, Sheets MD (2022) Bicaudal-C Post-transcriptional regulator of cell fates and functions. *Front Cell Dev Biol* **10** https://doi.org/10.3389/fcell.2022.981696

27. **Wessely O,** Tran U, Zakin L, De Robertis EM (2001) Identification and expression of the mammalian homologue of Bicaudal-C. *Mech Dev* **101**:267-270

28. **Wessely O,** De Robertis EM (2000) The Xenopus homologue of Bicaudal-C is a localized maternal mRNA that can induce endoderm formation. *Development* **127**:2053-2062

29. **Mahone M,** Saffman EE, Lasko PF (1995) Localized *Bicaudal-C* RNA encodes a protein containing a KH domain, the RNA binding motif of FMR1. *EMBO J* 14

30. **Rothe B,** Ikawa Y, Zhang Z, et al. (2023) Bicc1 ribonucleoprotein complexes specifying organ laterality are licensed by ANKS6-induced structural remodeling of associated ANKS3. *PLoS Biol* **21**:e3002302 https://doi.org/10.1371/journal.pbio.3002302

31. **Gamberi C,** Lasko P (2012) The Bic-C family of developmental translational regulators. *Comp Funct Genomics* **141386** https://doi.org/10.1155/2012/141386

32. **Piazzon N,** Maisonneuve C, Guilleret I, Rotman S, Constam DB (2012) Bicc1 links the regulation of cAMP signaling in polycystic kidneys to microRNA-induced gene silencing. *Journal of molecular cell biology* https://doi.org/10.1093/jmcb/mjs027

33. **Kilkenny C,** Browne WJ, Cuthill IC, Emerson M, Altman DG (2010) Improving bioscience research reporting: the ARRIVE guidelines for reporting animal research. *PLoS Biol* **8**:e1000412 https://doi.org/10.1371/journal.pbio.1000412

34. **Lian P,** Li A, Li Y, et al. (2014) Loss of polycystin-1 inhibits Bicc1 expression during mouse development. *PLoS One* **9**:e88816 https://doi.org/10.1371/journal.pone.0088816

35. **Lemaire LA,** Goulley J, Kim YH, et al. (2015) Bicaudal C1 promotes pancreatic NEUROG3+ endocrine progenitor differentiation and ductal morphogenesis. *Development* **142**:858-870 https://doi.org/10.1242/dev.114611

36. **Mesner LD,** Ray B, Hsu YH, et al. (2014) Bicc1 is a genetic determinant of osteoblastogenesis and bone mineral density. *J Clin Invest* **124**:2736-2749 https://doi.org/10.1172/JCI73072

37. **Xu Y,** Streets AJ, Hounslow AM, et al. (2016) The Polycystin-1, Lipoxygenase, and alpha-Toxin Domain Regulates Polycystin-1 Trafficking. *J Am Soc Nephrol* **27**:1159-1173 https://doi.org/10.1681/ASN.2014111074

38. **Xu Y,** Streets AJ, Hounslow AM, et al. (2015) The Polycystin-1, Lipoxygenase, and alpha-Toxin Domain Regulates Polycystin-1 Trafficking. *J Am Soc Nephrol* https://doi.org/10.1681/ASN.2014111074

39. **Naert T,** Cicek O, Ogar P, et al. (2021) Deep learning is widely applicable to phenotyping embryonic development and disease. *Development* **148** https://doi.org/10.1242/dev.199664

40. **Wu G,** Markowitz GS, Li L, et al. (2000) Cardiac defects and renal failure in mice with targeted mutations in Pkd2. *Nat Genet* **24**:75-78

41. **Wu G,** Tian X, Nishimura S, et al. (2002) Trans-heterozygous Pkd1 and Pkd2 mutations modify expression of polycystic kidney disease. *Hum Mol Genet* **11**:1845-1854

42. **Nauta J,** Ozawa Y, Sweeney WE, Rutledge JC, Avner ED. (1993) Renal and biliary abnormalities in a new murine model of autosomal recessive polycystic kidney disease. *Pediatr Nephrol* **7**:163-172

43. **Williams SS,** Cobo-Stark P, Hajarnis S, et al. (2014) Tissue-specific regulation of the mouse Pkhd1 (ARPKD) gene promoter. *Am J Physiol Renal Physiol* **307**:F356-368 https://doi.org/10.1152/ajprenal.00422.2013

44. **Devane J,** Ott E, Olinger EG, et al. (2022) Progressive liver, kidney, and heart degeneration in children and adults affected by TULP3 mutations. *Am J Hum Genet* **109**:928-943 https://doi.org/10.1016/j.ajhg.2022.03.015

45. **Lu H,** Galeano MCR, Ott E, et al. (2017) Mutations in DZIP1L, which encodes a ciliary-transition-zone protein, cause autosomal recessive polycystic kidney disease. *Nat Genet* **49**:1025-1034 https://doi.org/10.1038/ng.3871

46. **Magistroni R,** He N, Wang K, et al. (2003) Genotype-renal function correlation in type 2 autosomal dominant polycystic kidney disease. *Jasn* **14**:1164-1174

47. **Feng S,** Rodat-Despoix L, Delmas P, Ong AC (2011) A single amino acid residue constitutes the third dimerization domain essential for the assembly and function of the tetrameric polycystin-2 (TRPP2) channel. *J Biol Chem* **286**:18994-19000 https://doi.org/10.1074/jbc.M110.192286

48. **Dedoussis GV,** Luo Y, Starremans P, et al. (2008) Co-inheritance of a PKD1 mutation and homozygous PKD2 variant: a potential modifier in autosomal dominant polycystic kidney disease. *Eur J Clin Invest* **38**:180-190 https://doi.org/10.1111/j.1365-2362.2007.01913.x

49. **Stagner EE,** Bouvrette DJ, Cheng J, Bryda EC (2009) The polycystic kidney disease-related proteins Bicc1 and SamCystin interact. *Biochem Biophys Res Commun* **383**:16-21 https://doi.org/10.1016/j.bbrc.2009.03.113

50. **Kim DY,** Woo YM, Lee S, et al. (2019) Impact of miR-192 and miR-194 on cyst enlargement through EMT in autosomal dominant polycystic kidney disease. *FASEB J* **33**:2870-2884 https://doi.org/10.1096/fj.201800563RR

51. **Formica C,** Peters DJM (2020) Molecular pathways involved in injury-repair and ADPKD progression. *Cell Signal* **72** https://doi.org/10.1016/j.cellsig.2020.109648

52. **Milutinovic J,** Rust PF, Fialkow PJ, et al. (1992) Intrafamilial phenotypic expression of autosomal dominant polycystic kidney disease. *American journal of kidney diseases: the official journal of the National Kidney Foundation* **19**:465-472

53. **Harris PC,** Rossetti S (2010) Determinants of renal disease variability in ADPKD. Review. *Advances in chronic kidney disease* **17**:131-139 https://doi.org/10.1053/j.ackd.2009.12.004

54. **Minegishi K,** Rothe B, Komatsu KR, et al. (2021) Fluid flow-induced left-right asymmetric decay of Dand5 mRNA in the mouse embryo requires a Bicc1-Ccr4 RNA degradation complex. *Nat Commun* **12**:4071 https://doi.org/10.1038/s41467-021-24295-2

55. **Maerker M,** Getwan M, Dowdle ME, et al. (2021) Bicc1 and Dicer regulate left-right patterning through post-transcriptional control of the Nodal inhibitor Dand5. *Nat Commun* **12**:5482 https://doi.org/10.1038/s41467-021-25464-z

56. **Wang L,** Eckmann CR, Kadyk LC, Wickens M, Kimble J (2002) A regulatory cytoplasmic poly(A) polymerase in Caenorhabditis elegans. *Nature* **419**:312-316

57. **Chicoine J,** Benoit P, Gamberi C, Paliouras M, Simonelig M, Lasko P (2007) Bicaudal-C recruits CCR4-NOT deadenylase to target mRNAs and regulates oogenesis, cytoskeletal organization, and its own expression. *Dev Cell* **13**:691-704 https://doi.org/10.1016/j.devcel.2007.10.002

58. **Zhang Y,** Park S, Blaser S, Sheets MD (2014) Determinants of RNA binding and translational repression by the Bicaudal-C regulatory protein. *J Biol Chem* **289**:7497-7504 https://doi.org/10.1074/jbc.M113.526426

59. **Zhang Y,** Cooke A, Park S, Dewey CN, Wickens M, Sheets MD (2013) Bicaudal-C spatially controls translation of vertebrate maternal mRNAs. *RNA* **19**:1575-1582 https://doi.org/10.1261/rna.041665.113

60. **Chu AS,** Friedman JR (2008) A role for microRNA in cystic liver and kidney diseases. *J Clin Invest* **118**:3585-3587 https://doi.org/10.1172/JCI36870

61. **Patel V,** Williams D, Hajarnis S, et al. (2013) miR-17~92 miRNA cluster promotes kidney cyst growth in polycystic kidney disease. *Proc Natl Acad Sci USA* **110**:10765-10770 https://doi.org/10.1073/pnas.1301693110

62. **Pandey P,** Brors B, Srivastava PK, et al. (2008) Microarray-based approach identifies microRNAs and their target functional patterns in polycystic kidney disease. *BMC Genomics* **9** https://doi.org/10.1186/1471-2164-9-624

63. **Pandey P,** Qin S, Ho J, Zhou J, Kreidberg JA (2011) Systems biology approach to identify transcriptome reprogramming and candidate microRNA targets during the progression of polycystic kidney disease. *BMC systems biology* **5** https://doi.org/10.1186/1752-0509-5-56

64. **Patel V,** Hajarnis S, Williams D, Hunter R, Huynh D, Igarashi P (2012) MicroRNAs regulate renal tubule maturation through modulation of Pkd1. *J Am Soc Nephrol* **23**:1941-1948 https://doi.org/10.1681/ASN.2012030321

65. **Nagalakshmi VK,** Ren Q, Pugh MM, Valerius MT, McMahon AP, Yu J (2011) Dicer regulates the development of nephrogenic and ureteric compartments in the mammalian kidney. *Kidney Int* **79**:317-330 https://doi.org/10.1038/ki.2010.385

66. **Yheskel M,** Lakhia R, Cobo-Stark P, Flaten A, Patel V (2019) Anti-microRNA screen uncovers miR-17 family within miR-17~92 cluster as the primary driver of kidney cyst growth. *Sci Rep* **9**:1920 https://doi.org/10.1038/s41598-019-38566-y

67. **Lakhia R,** Ramalingam H, Chang CM, et al. (2022) PKD1 and PKD2 mRNA cis-inhibition drives polycystic kidney disease progression. *Nat Commun* **13**:4765 https://doi.org/10.1038/s41467-022-32543-2

68. **Lee EC**, Valencia T, Allerson C, et al. (2019) Discovery and preclinical evaluation of anti-miR-17 oligonucleotide RGLS4326 for the treatment of polycystic kidney disease. *Nat Commun* **10**:4148 https://doi.org/10.1038/s41467-019-11918-y

Peer reviews

Reviewer #1 (Public review):

In this manuscript, Tran et al. investigate the interaction between BICC1 and ADPKD genes in renal cystogenesis. Using biochemical approaches, they reveal a physical association between Bicc1 and PC1 or PC2 and identify the motifs in each protein required for binding. Through genetic analyses, they demonstrate that Bicc1 inactivation synergizes with Pkd1 or Pkd2 inactivation to exacerbate PKD-associated phenotypes in Xenopus embryos and potentially in mouse models. Furthermore, by analyzing a large cohort of PKD patients, the authors identify compound BICC1 variants alongside PKD1 or PKD2 variants in trans, as well as homozygous BICC1 variants in patients with early-onset and severe disease presentation. They also show that these BICC1 variants repress PC2 expression in cultured cells.

Overall, the concept that BICC1 variants modify PKD severity is plausible, the data are robust, and the conclusions are largely supported.

Comments on revision:

My comments have been mostly addressed.

https://doi.org/10.7554/eLife.106342.2.sa3

Reviewer #2 (Public review):

Tran and colleagues report evidence supporting the expected yet undemonstrated interaction between the Pkd1 and Pkd2 gene products Pc1 and Pc2 and the Bicc1 protein in vitro, in mice, and collaterally, in Xenopus and HEK293T cells. The authors go on to convincingly identify two large and non-overlapping regions of the Bicc1 protein important for each interaction and to perform gene dosage experiments in mice that suggest that Bicc1 loss of function may compound with Pkd1 and Pkd2 decreased function, resulting in PKD-like renal phenotypes of different severity. These results led to examining a cohort of very early onset PKD patients to find three instances of co-existing mutations in PKD1 (or PKD2) and BICC1. Finally, preliminary transcriptomics of edited lines gave variable and subtle differences that align with the theme that Bicc1 may contribute to the PKD defects, yet are mechanistically inconclusive.

These results are potentially interesting, despite the limitation, also recognized by the authors, that BICC1 mutations seem exceedingly rare in PKD patients and may not "significantly contribute to the mutational load in ADPKD or ARPKD". The manuscript has several intrinsic limitations that must be addressed.

The manuscript contains factual errors, imprecisions, and language ambiguities. This has the effect of making this reviewer wonder how thorough the research reported and analyses have been.

Comments on revision:

My comments have been addressed.

https://doi.org/10.7554/eLife.106342.2.sa2

Reviewer #3 (Public review):

Summary:

This study investigates the role of BICC1 in the regulation of PKD1 and PKD2 and its impact on cytogenesis in ADPKD. By utilizing co-IP and functional assays, the authors demonstrate physical, functional, and regulatory interactions between these three proteins.

Strengths:

(1) The scientific principles and methodology adopted in this study are excellent, logical, and reveal important insights into the molecular basis of cystogenesis.

(2) The functional studies in animal models provide tantalizing data that may lead to a further understanding and may consequently lead to the ultimate goal of finding a molecular therapy for this incurable condition.

(3) In describing the patients from the Arab cohort, the authors have provided excellent human data for further investigation in large ADPKD cohorts. Even though there was no patient material available, such as HUREC, the authors have studied the effects of BICC1 mutations and demonstrated its functional importance in a Xenopus model.

Weaknesses:

This is a well-conducted study and could have been even more impactful if primary patient material was available to the authors. A further study in HUREC cells investigating the critical regulatory role of BICC1 and potential interaction with mir-17 may yet lead to a modifiable therapeutic target.

Conclusion:

The authors achieve their aims. The results reliably demonstrate the physical and functional interaction between BICC1 and PKD1/PKD2 genes and their products.

The impact is hopefully going to be manifold:

(1) Progressing the understanding of the regulation of the expression of PKD1/PKD2 genes.

Comments on revision:

My comments have been addressed and sorted.

https://doi.org/10.7554/eLife.106342.2.sa1

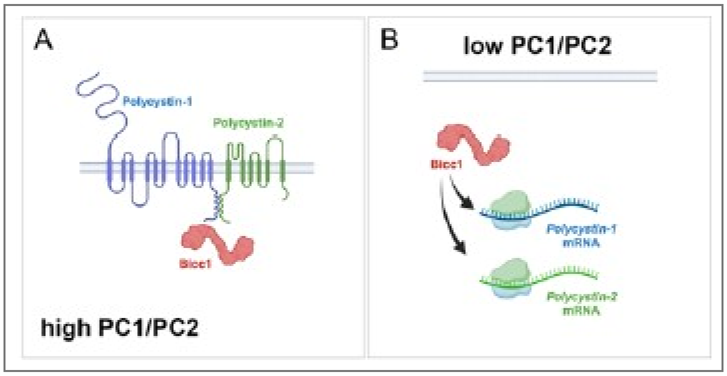
Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *(1) The authors devote significant effort to characterizing the physical interaction between Bicc1 and Pkd2. However, the study does not examine or discuss how this interaction relates to Bicc1's well-established role in posttranscriptional regulation of Pkd2 mRNA stability and translation efficiency.*

The reviewer is correct that the present study has not addressed the downstream consequences of uthis interaction considering that Bicc1 is a posttranscriptional regulator of Pkd2 (and potentially Pkd1). We think that the complex of Bicc1/Pkd1/Pkd2 retains Bicc1 in the cytoplasm and thus restrict its activity in participating in posttranscriptional regulation (see Author response image 1). We, however, do not yet have data to support this and thus have not included this model in the manuscript. Yet, we have updated the discussion of the manuscript to further elaborate on the potential mechanism of the Bicc1/Pkd1/Pkd2 complex.

We have updated the discussion to include a discussion on the potential consequences on posttranscriptional regulation by Bicc1.



Author response image 1. Model of BICC1, PC1 and PC2 self-regulation. In this model Bicc1 acts as a positive regulator of PKD gene expression. In the presence of 'sufficient' amounts of PC1/PC2 complex, it is tethered to the complex and remains biologically inactive (Fig. 1A). However, once the levels of the PC1/PC2 complex are reduced, Bicc1 is now present in the cytoplasm to promote expression of the PKD proteins, thereby raising their levels (Fig. 4B), which then in turn will 'shutdown' Bicc1 activity by again tethering it to the plasma membrane.

> *(2) Bicc1 inactivation appears to downregulate Pkd1 expression, yet it remains unclear whether Bicc1 regulates Pkd1 through direct interaction or by antagonizing miR-17, as observed in Pkd2 regulation. This should be further examined or discussed.*

This is a very interesting comment. Vishal Patel published that PKD1 is regulated by a mir-17 binding site in its 3'UTR (PMID: 35965273). We, however, have not evaluated whether BICC1 participates in this regulation. A definitive answer would require utilization of the mice described in above reference, which is beyond the scope of this manuscript. We, however, have revised the discussion to elaborate on this potential mechanism.

We have updated the discussion to include a statement on the potential direct regulation of Pkd1 mRNA by Bicc1.

> *(3) The evidence supporting Bicc1 and ADPKD gene cooperativity, particularly with Pkd1, in mouse models is not entirely convincing, likely due to substantial variability and the aggressive nature of Bpk/Bpk mice. Increasing the number of animals or using a milder Bicc1 strain, such as jcpk heterozygotes, could help substantiate the genetic interaction.*

We have initially performed the analysis using our Bicc1 complete knockout, we previously reported on (PMID 20215348) focusing on compound heterozygotes. Yet, similar to the Pkd1/Pkd2 compound heterozygotes (PMID 12140187) no cyst development was observed when we sacrificed the mice as late as P21. Our strain is similar to the above mentioned jcpk, which is characterized by a short, abnormal transcript thought to result in a null allele (PMID: 12682776). We thank the reviewer for pointing us to the reference showing the heterozygous mice exhibit glomerular cysts in the adults (PMID: 7723240). This suggestion is an interesting idea we will investigate. In general, we agree with the reviewer that a better understanding of the contribution of Bicc1 to the adult PKD phenotype will be critical. To this end, we are currently generating a floxed allele of Bicc1 that will allow us to address the cooperativity in the adult kidney, when e.g. crossed to the $Pkd1^{RC/RC}$ mice. Yet, these experiments are beyond the timeframe for this revision.

No changes were made in the revised manuscript.

> ***Reviewer #2 (Public review):***
>
> *(1) These results are potentially interesting, despite the limitation, also recognized by the authors, that BICC1 mutations seem exceedingly rare in PKD patients and may not "significantly contribute to the mutational load in ADPKD or ARPKD". The manuscript has several intrinsic limitations that must be addressed.*

As mentioned above, the study was designed to explore whether there is an interaction between BICC1 and the PKD1/PKD2 and whether this interaction is functionally important. How this translates into the clinical relevance will require additional studies (and we have addressed this in the discussion of the manuscript).

> *(2) The manuscript contains factual errors, imprecisions, and language ambiguities. This has the effect of making this reviewer wonder how thorough the research reported and analyses have been.*

We respectfully disagree with the reviewer on the latter interpretation. The study was performed with rigor. We have carefully assessed the critiques raised by the reviewer. As presented below, most of the criticisms raised by the reviewer have been easily addressed in the revised version of the manuscript. Yet, none of the critiques seems to directly impact the overall interpretation of the data.

> ***Reviewer #1 (Recommendations for the authors):***
>
> *(1) The manuscript requires further editing. For example, figure panels and legends are mismatched in Figure 1*

We have corrected the labeling of Figure 1.

> *(2) Y-axis units and values are inconsistent in Figures 4b-4g, Supplementary Figures S2e and S2f are not referenced in the text, genotypes are missing in Supplementary Figure S3f, and numerous typographical errors are present.*

In respect to the y-axis in Figure 4b-g, the scale is different for each of them, but that is intentional as one would lose the differences if they were all scaled identically. But we have now mentioned this in the figure legend to make the reader aware of it. In respect to the Supplemental Figure S2e,f, we included the panels in the description of the mutant BICC1 lines, but unfortunately forgot to reference them. This has now been done.

We have updated the labeling of the Y-axis for the cystic indices adding “[%]” as the unit and updated the figure legend of Figure 4. We have included the genotypes in Supplementary Figure S3f. The Supplementary Figure S2e,f is now mentioned in the supplemental material (page 9, 2^{nd} paragraph).

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) Previous data from mouse, Xenopus, and zebrafish suggest a crucial role for the RNAbinding protein Bicc1 in the pathogenesis of PKD, although BICC1 mutations in human PKD have not been previously reported." The cited sources (and others that were not cited) link Bicc1 mutations to renal cysts, similar to a report by Kraus (PMID: 21922595) that the authors cite later. However, a more direct link to PKD was reported by Lian and colleagues using whole Pkd1 mice (PMID: 20219263) and by Gamberi and*

> *colleagues using Pkd1 kidneys and human microarrays (PMID: 28406902). Although relevant, neither is cited here, and only the former is cited later in the manuscript.*

Thanks for pointing this out. We have added these three citations.

We have added these three citations (PMID: 21922595, PMID: 20219263 and PMID: 28406902) in the indicated sentence.

> *(2) In Figure 1B, the lanes do not seem to correspond among panels, particularly evident in the panel with myc-mBicc1. Hence, it is difficult to agree with the presented conclusions.*

We have corrected the labeling of the lanes in Figure 1b.

> *(3) In the Figure 1 legend: "(g) Western blot analysis following co-IP experiments, using an anti-mouse Bicc1 or anti-goat PC2 antibody as bait, identified protein interactions between endogenous PC2 and BICC1 in UCL93 cells. Non-immune goat and mouse IgG were included as a negative control." There is no mention of panel H, although this reviewer can imagine what the authors meant. The capitalization differs in the figure and legend. More troublingly, in panel G, a non-defined star indicates a strong band present in both immune and non-immune control.*

We have corrected the figure legend of Figure 1 and clarified the non-specific band in the figure legend.

> *(4) In Figure 4, the authors do not show the matched control for the Bicc1 Pkd1 interaction in panel d, nor do they show a scale bar in either a) or d). Thus, the phenotypic severity cannot be properly assessed.*

Thanks for pointing out the missing scale bars, which have now been added. In respect to the two kidneys shown in Figure 4d, the two kidneys shown are from littermates to illustrate the kidney size in agreement with the cumulative data shown in Figure 4e. Unfortunately, this litter did not have a wildtype control. As the data analysis in Figure 4e is based on littermates, mixing and matching kidneys of different litters does not seem appropriate. Thus, we have omitted showing a wildtype control in this panel. However, the size of the wildtype kidney can be seen in Figure 4a.

We have added the scale bar to both panels and have updated the figure legend to emphasize that the kidneys shown are from littermates and that no wildtype littermate was present in this litter.

> *(5) "Surprisingly, an 8-fold stronger interaction was observed between full-length PC1 and myc-mBicc1-ΔKH compared to mycmBicc1 or myc-mBicc1-ΔSAM." Assuming all the controls for protein folding and expression levels have been carried out and not shown/mentioned, this sentence seems to contradict the previous statement that Bicc1deltaSAM reduced the interaction with PC1 by 55%. Because the full length and SAM deletion have different interaction strengths, the latter sentence makes no sense.*

The reduction in the levels of myc-mBicc1-ΔSAM compared to wildtype mycmBicc1 in respect to PC1 binding was not significant. We have clarified this in the text.

We have corrected the sentence and modified the Figure accordingly.

> *(6) Imprecise statements make a reader wonder how to interpret the data: "More than three independent experiments were analyzed." Stating the sample size or including it in the figure would save space and improve confidence in the data presented.*

We have stated the exact number of animals per conditions above each of the bars.

> *(7) "Next, we performed a similar mouse study for Pkd1 by reducing the gene dose of Pkd1 postnatally in the collecting ducts using a Pkhd1-Cre as previously described40" What did the authors mean?*

The reference was included to cite the mouse strain, but realized that it can be misinterpreted that the exact experiments has been performed previously. We have clarified this in the text.

We have reworded the sentence to avoid misinterpretation.

> *(8) The authors examined the additive effects of knocking down Bicc1, Pkd1, and Pkd2 with morpholinos in Xenopus and, genetically, in mice. While the Bicc1[+/-] Pkd1 or 2[+/-] double heterozygote mice did not show phenotypes, the authors report that the Bicc1[-/-] Pkd1 or 2 [+/-] did instead show enlarged kidneys. What is the phenotype of a Bicc1[+/-] Pkd1 or 2 [-/-]? What we learn from the author's findings among the PKD population suggests that the latter situation would be potentially translationally relevant.*

The mouse experiments were designed to address a cooperativity between Bicc1 and either Pkd1 or Pkd2 and whether removal of one copy of Pkd1 or Pkd2 would further worsen the Bicc1 cystic kidney phenotype. Thus, the parental crosses were chosen to maximize the number of animals obtained for these genotypes. Unfortunately, these crosses did not yield the genotypes requested by the reviewer. To address the contribution of Bicc1 towards the PKD population, we will need to perform a different cross, where we eliminate Pkd1 or Pkd2 in a floxed background of Bicc1 postnatally in adult mice. While we are gearing up to perform such an experiment, this is timewise beyond the scope of the manuscript. In addition, please note that we have addressed the question about the translation towards the PKD population already in the discussion of the original submission (page 13/14, last/first paragraph).

No changes have been made to the revised version of the manuscript.

> *(9) How do the authors interpret the milder effects of the Bicc1[-/-] Pkd1[+/-] compared to Bicc1[-/-] Pkd2[+/-] relative to the respective protein-protein interactions?*

The milder effects are due to the nature of the crosses. While the Pkd2 mutant is a germline mutation, the Pkd1 mutant is a conditional allele eliminating Pkd1 only in the collecting ducts of the kidney. As such, we spare other nephron segments such as the proximal tubules, which also significantly contribute to the cyst load. As such these mouse data support the interaction between Pkd1 and Pkd2 with Bicc1, but do not allow us to directly compare the outcomes. While this was mentioned in the previous version of the manuscript, we have expanded on this in the revised version of the manuscript.

We have expanded the results section in the revised version of the manuscript highlighting that the two different approaches cannot be directly compared.

> *(10) How do the authors interpret that the strong Bicc1[Bpk] Pkd1 or Pkd2 double heterozygote mice did not have defects and "kidneys from Bicc1+/-:Pkd2+/- did not exhibit cysts (data not shown)", when the VEO PKD patients and - although not a genetic reduction - also the morpholino-treated Xenopus did?*

VEO PKD patients are characterized by a loss of function of PKD1 or PKD2 and – as we propose in this manuscript - that BICC1 further aggravates the phenotype. Yet, we do not address either in the mouse or Xenopus experiments whether BICC1 is a genetic modifier. We are simply addressing whether the two genes show a genetic interaction. In the mouse studies, we eliminate one copy of Pkd1 or Pkd2 in the background of a hypomorphic allele of Bicc1. Similarly, in the Xenopus experiments, we employ suboptimal doses of the morpholino oligomers, i.e., concentrations that did not yield a phenotypic change and then asked whether removing both together show cooperativity. It is important to state that this is based on a biological readout and not defined based on the amount of protein. While we have described this already in the original manuscript (page 7, first paragraph), we have amended our description of the Xenopus experiment to make this even clearer.

Finally, we agree with the reviewer that if we were to address whether Bicc1 is a modifier of the PKD phenotype in mouse, we would need to reduce Bicc1 function in a Pkd1 or Pkd2 mutants. Yet, we have recognized this already in the initial version of the manuscript in the discussion (page 14, first paragraph).

We have expanded the results section when discussing the suboptimal amounts of the morpholino oligos (Page 6, 1st paragraph).

> *(11) Unclear: "While variants in BICC1 are very rare, we could identify two patients with BICC1 variants harboring an additional PKD2 or PKD1 variant in trans, respectively." Shortly after, the authors state in apparent contradiction that "the patients had no other variants in any of other PKD genes or genes which phenocopy PKD including PKD1, PKD2, PKHD1, HNF1s, GANAB, IFT140, DZIP1L, CYS1, DNAJB11, ALG5, ALG8, ALG9, LRP5, NEK8, OFD1, or PMM2."*

The reviewer is correct. This should have been phrased differently. We have now added “Besides the variants reported below” to clarify this more adequately.

The sentence was changed to start with “Besides the variants reported below, […].”

> *(12) "The demonstrated interaction of BICC1, PC1, and PC2 now provides a molecular mechanism that can explain some of the phenotypic variability in these families." How do the authors reconcile this statement with their reported ultra-rare occurrence of the BICC1 mutations?*

As mentioned in the manuscript and also in response to the other two reviewers, Bicc1 has been shown to regulate Pkd2 gene expression in mice and frogs via an interaction with the miR-17 family of microRNAs. Moreover, the miR-17 family has been demonstrated to be critical in PKD (PMID: 30760828, PMID: 35965273, PMID: 31515477, PMID: 30760828). In fact, both other reviewers have pointed out that we should stress this more since Bicc1 is part of this regulatory pathway. Future experiments are needed to address whether Bicc1 contributes to the variability in ADPKD onset/severity. Yet, this is beyond the scope of this study.

Based on the comments of the two other reviewers we have further addressed the Bicc1/miR-17 interaction.

> *(13) The manuscript should use correct genetic conventions of italicization and capitalization. This is an issue affecting the entire manuscript. Some exemplary instances are listed below.*
>
> *(a) "We also demonstrate that Pkd1 and Pkd2 modifies the cystic phenotype in Bicc1 mice in a dose-dependent manner and that Bicc1 functionally interacts with Pkd1, Pkd2 and Pkhd1 in the pronephros of Xenopus embryos." Genes? Proteins?*

The data presented in this section show that a hypomorphic allele of Bicc1 in mouse and a knockdown in Xenopus yields this. As both affect the proteins, the spelling should reflect the proteins.

No changes have been made in the revised manuscript.

> *(b) The sentence seems to use both the human and mouse genetic capitalization, although it refers to experiments in the mouse system "to define the Bicc1 interacting domains for PC2 (Fig. 2d,e). Full-length PC2 (PC2-HA) interacted with full-length myc-mBICC1."*

We agree with the review that stating the species of the molecules used is critical, we have adapted a spelling of Bicc1, where BICC1 is the human homologue, mBicc1 is the mouse homologue and xBicc1 the Xenopus one.

We have highlighted the species spelling in the methods section and labeled the species accordingly throughout the manuscript and figures.

> *(14) "Together these data supported our biochemical interaction data and demonstrated that BICC1 cooperated with PKD1 and PKD2." Are the authors implying that these results in mice will translate to the human protein?*

We agree that we have not formally shown that the same applies to the human proteins. Thus, we have changed the spelling accordingly.

We have revised the capitalization of the proteins.

> *(15) The text is often unclear, terse, or inconsistent.*
>
> *(a) "These results suggested that the interaction between PC1 and Bicc1 involves the SAM but not the KH/KHL domains (or the first 132 amino acids of Bicc1). It also suggests that the N-terminus could have an inhibitory effect on PC1-BICC1 association." How do the authors define the N-terminus? The first 132 aa? KH/KHL domains?*

This was illustrated in the original Figure 2A. The DKH constructs lack the first 351 amino acids.

To make this more evident, we have specified this in the text as well.

> *(b) Similarly, the authors state below, "Unlike PC1, PC2 interacted with mycmBICC1ΔSAM, but not myc-mBICC1-ΔKH suggesting that PC2 binding is dependent on the N-terminal domains but not the SAM domain." It is unclear if the authors refer to the KH/KHL domains or others. Whatever the reference to the N-terminal region, it should also be consistent with the section above.*

This is now specified in the text.

> *(c) Unclear: "We have previously demonstrated that Pkd2 levels are reduced in a complete Bicc1 null mice,22 performing qRT-PCR of P4 kidneys (i.e. before the onset of a strong cystic phenotype), revealed that Bicc1, Pkd1 and Pkd2 were statistically significantly down9 regulated (Fig. 4h-j)".*

We have changed the text to clarify this.

> *(d) "Utilizing recombinant GST domains of PC1 and PC2, we demonstrated that BICC1 binds to both proteins in GST-pulldown assays (Fig. 1a, b)." GST-tagged domains? Fusions?*

We have changed the text to clarify this.

> *(e) "To study the interaction between BICC1, PKD1 and PKD2 we combined biochemical approaches, knockout studies in mice and Xenopus, genetic engineered human kidney cells" > genetically engineered.*

We have changed the text to clarify this.

> *(f) Capitalization (e.g., see Figure S3, ref. the Bpk allele) and annotation (e.g., Gly821Glu and G821E) are inconsistent.*

We have homogenized the labeling of the capitalization and annotations throughout the manuscript.

> *(g) What do the authors mean by "homozygous evolutionarily well-conserved missense variant"?*

We have changed this is the revised version of the manuscript.

> ***Reviewer #3 (Public review/Recommendations to the authors):***
>
> *(1) A further study in HUREC cells investigating the critical regulatory role of BICC1 and potential interaction with mir-17 may yet lead to a modifiable therapeutic target.*
>
> *(2) This study should ideally include experiments in HUREC material obtained from patients/families with BICC1 mutations and studying its effects on the PKD1/2 complex in primary cell lines.*

This is an excellent suggestion. We agree with the reviewer that it would have been interesting to analyze HUREC material from the affected patients. Unfortunately, besides DNA and the phenotypic analysis described in the manuscript neither human tissue nor primary patient-derived cells collected once the two patients with the BICC1 p.Ser240Pro variant passed away.

No changes to the revised manuscript have been made to address this point.

> *(3) Please remove repeated words in the following sentence in paragraph 2 of the introduction: "BICC1 encodes an evolutionarily conserved protein that is characterized by 3 K-homology (KH) and 2 KH-like (KHL) RNA-binding domains at the N-terminus and a SAM domain at the C-terminus, which are separated by a by a disordered intervening sequence (IVS).23-28".*

This has been changed.

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